

BiblioBlitz

Q1 Open-Access Academic Paper Downloader

Version 3.0 | Pure Python | Cross-Platform

1. About BiblioBlitz

BiblioBlitz is a desktop application for researchers and students who need to collect open-access academic papers in bulk. It searches high-impact, Q1-ranked journals via the CrossRef API, checks each paper for a freely available PDF through Unpaywall, and downloads the files automatically — all from a clean graphical interface.

The tool is aimed at fields such as hydrology, climate science, earth sciences, environmental science, remote sensing, and ecology, though its keyword-based search makes it useful for any discipline covered by the included journal list.

2. Version Details

BiblioBlitz v3.0 (current release)

Version 3.0 is a complete rewrite of the earlier PaperFinder (v2) tool. The backend has been moved from R and PowerShell to pure Python, which means the application runs on any operating system and architecture. End users do not need R or any scientific Python library installed.

Feature	v2 — PaperFinder	v3 — BiblioBlitz
Backend	R + PowerShell	Pure Python
OS support	Windows 64-bit only	Any OS, any architecture
Max papers	10,000	1,00,000
Journal filter	All journals	Q1 journals only
Input placeholders	Hard-coded defaults	Smart (auto-clear on type)
Logo display	File-path dependent	Always visible (embedded)

3. Python Libraries Used

BiblioBlitz uses only lightweight Python libraries. End users running the pre-built EXE do not need to install anything.

- **customtkinter** — Provides the modern graphical user interface. Built on top of tkinter with improved, cleaner-looking widgets.
- **tkinter** (standard library) — The base GUI framework included with Python. Handles file dialogs, message boxes, and widget infrastructure.

- **Pillow (PIL)** — Handles image loading and rendering inside the application window, including the embedded app logo.
- **urllib** (standard library) — Performs all HTTP requests: querying the CrossRef and Unpaywall APIs and downloading PDF files.
- **threading** (standard library) — Runs the download process in a background thread so the interface stays responsive. Allows stopping at any time.
- **json, csv, re, os, pathlib** (standard library) — Handle API responses, write the download log, clean file names, and manage directories.
- **PyInstaller** (build time only) — Packages the script and all dependencies into a single standalone EXE for Windows.

4. What BiblioBlitz Does

BiblioBlitz automates finding and downloading open-access research papers from top-ranked journals. Given keywords and a year range, it searches CrossRef, filters to Q1 journals, checks Unpaywall for free PDFs, and saves everything to a folder you choose along with a summary CSV log.

5. How to Use BiblioBlitz

Step 1 — Get the Application

On Windows: make sure Python 3.9 or newer is installed, then double-click `build_exe.bat`. It installs required packages and produces `dist\BiblioBlitz.exe`. No R needed, no extra setup.

On macOS or Linux: run the script from a terminal with: `python biblioblitz.py`

Step 2 — Fill in the Fields

- **Email address:** Your institutional or personal email. Sent in the API request headers so CrossRef and Unpaywall can reach you if needed. Not stored or shared.
- **Search keywords:** One or more keywords separated by commas (e.g. hydrology, streamflow, flood). Broader terms give more results.
- **Download directory:** Click Browse to choose where PDFs are saved. Created automatically if it does not exist.
- **Max papers:** How many papers to attempt (1 to 1,00,000). Larger runs take more time.
- **Starting year:** Only papers from this year onward are included. Range: 1990 to 2025.

Step 3 — Start the Download

Click Start Download. The log panel shows progress in real time — papers found, Q1 filter results, open-access count, downloaded count. Click Stop at any time; the current file finishes before stopping.

Step 4 — Check the Results

Your folder will contain PDFs named with the title and DOI. A `download_log.csv` file records every paper processed: title, DOI, journal, year, open-access status, and result.

Step 5 — PDF Integrity Check (Optional)

Click Run PDF Integrity Check after downloading. The tool verifies each PDF and moves any corrupt or incomplete files to a quarantine subfolder.

6. Q1 Journal Coverage

BiblioBlitz restricts searches to Q1-ranked journals. The built-in list covers journals from these publishers:

- **Nature family** — Nature, Nature Climate Change, Nature Geoscience, Nature Water, Nature Communications, Scientific Reports, and more.
- **Science / AAAS** — Science, Science Advances.
- **EGU / Copernicus** — HESS, NHESS, The Cryosphere, Atmospheric Chemistry and Physics, Biogeosciences, and more.
- **AGU / Wiley** — Geophysical Research Letters, Journal of Geophysical Research (all series), Water Resources Research, and more.
- **Elsevier** — Journal of Hydrology, Advances in Water Resources, Science of the Total Environment, Remote Sensing of Environment, Catena, Geoderma, Geomorphology, and more.
- **AMS** — Journal of Climate, Journal of Hydrometeorology, Monthly Weather Review, Bulletin of the AMS.
- **Springer** — Climatic Change, Climate Dynamics, Hydrogeology Journal, Natural Hazards, Landslides, and others.
- **MDPI (Q1)** — Remote Sensing, Water, Atmosphere, Sustainability, Land.
- **Other high-impact** — PLOS ONE, Global Environmental Change, One Earth, iScience, Hydrological Sciences Journal, Ecohydrology.

7. Tips for Best Results

- Use broad keywords first (e.g. hydrology, climate change) for a wider set of results, then narrow with specific terms.
- Large downloads can take several hours. Consider running them overnight.
- CrossRef and Unpaywall are free and need no registration, but have rate limits. BiblioBlitz pauses automatically so you do not get blocked.
- Always run the PDF Integrity Check after a large download to remove files that did not transfer correctly.
- Review download_log.csv to see which papers had open-access PDFs and which were skipped.

8. Acknowledgements

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Claude (Anthropic) — Assisted with Python code structure, GUI design logic, debugging, and documentation drafting.

Gemini (Google) — Helped with the journal whitelist, API workflow design, and testing suggestions.

ChatGPT (OpenAI) — Contributed to early prototyping, code review, and general problem-solving.

The application also relies on two free services: CrossRef (crossref.org) for academic metadata, and Unpaywall (unpaywall.org) for open-access PDF links. Both are provided free of charge to the research community.

9. Contact

For questions, bug reports, or suggestions, please reach out to:

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